

Introduction

DEFINITIONS:

PHYSICS: Physics is the study of nature; in other words, it is the exploration of the fundamental principles which govern the behavior of the material universe.

There are many branches of physics:

Classical Mechanics

Classical Mechanics focuses on analyzing the movement of objects and the forces influencing them. Within its scope lie foundational ideas including Newton's laws of motion, the study of kinematics, the concept of momentum and the principles of energy.

Electromagnetism

Electromagnetic theory is the study of electric and magnetic fields, how they affect charged things, and how they work together. It's all about understanding electric and magnetic forces, how things like light travel and the basic ideas behind electricity and magnets.

Thermodynamics

Thermodynamics is the branch of science that deals with the energy transfer and the conversion of energy from one form to another.

Optics

Optics is the branch of physics that deals with the behavior of light, its interactions with matter, and the formation of images by optical instruments.

Quantum Mechanics:

Quantum mechanics is the fundamental theory of physics that describes the behavior of particles on the atomic and subatomic scale. It introduces the concept of wave functions, probabilities, and quantization of energy levels.

Astrophysics:

Astrophysics is the study of astronomical bodies, namely the planets, solar systems, galaxies, etc. It deals with the Physics involved in motion, formation and other phenomena related to astronomical bodies.

Plasma Physics:

Plasma physics is the study of ionized gases known as plasma, which are characterized by their unique properties, such as collective behavior, electromagnetic interactions, and relevance to controlled fusion energy.

Medical Physics:

Medical physics is the application of physics principles to the field of medicine, particularly in the areas of diagnostic imaging, radiation therapy, and radiation protection.

Nuclear Physics:

Nuclear physics is the study of the properties and behavior of atomic nuclei, encompassing the understanding of nuclear structure, reactions, and applications.

Physics XI

PHYSICS: SCOPE IN SCIENCE, TECHNOLOGY AND SOCIETY:

Physics is essential for understanding and shaping our world. It is the foundation of science, technology and society. Here are some key aspects highlighting the scope of physics in these areas:

SCIENCE	- Physics provides the fundamental laws and principles that govern the natural world from the smallest subatomic particles to the largest galaxies. - It drives scientific progress by pushing the boundaries of our understanding of the universe. - Physics intersects with other scientific disciplines such as chemistry, biology, astronomy, and geology, providing a framework for understanding complex systems and phenomena.
TECHNOLOGY	- Physics principles are employed in various engineering fields to design and develop new technologies. - Physics plays a crucial role in the generation, transmission, and utilization of energy. - Physics principles are fundamental to the development of electronic devices, telecommunications, and information technology.
SOCIETY	- Physics contributes to medical imaging technologies, radiation therapy, laser surgery, and medical diagnostics. - Physics research aids in understanding the properties of materials and developing new materials for various applications. - Physics plays a role in studying climate change, atmospheric physics, and environmental monitoring. - Physics education fosters critical thinking, problem-solving skills, and a scientific mindset, promoting scientific literacy and enabling individuals to make informed decisions on science-related topics.

In short, physics is a fundamental science that underpins our understanding of the natural world and enables technological advancements that benefit society.

INTERNATIONAL SYSTEM OF UNITS: (SI UNITS)

SI units (International system of units) are the system of units used for measurement internationally. Adopted in every aspect of study to ensure uniform and standardized measurements of physical quantities in every field.

These seven base SI units provide a fundamental framework for measuring various physical quantities and are used as building blocks to derive other SI units for more complex measurements.

- Meter (m):** The meter is the base unit for length or distance. It is defined as the distance that light travels in a vacuum in exactly $1/300$ millionths of a second.
- Kilogram (kg):** The kilogram is the base unit for mass. It is determined by the mass of a special metal block in France called the international prototype of the kilogram.
- Second (s):** The second is the base unit for time. It is defined as the duration of 9,192,631,770 periods of the radiation corresponding to the transition between two energy levels of the cesium-133 atom.

- Ampere (A):** The ampere is the base unit for electric current. It is defined as the current that, when flowing through two infinitely long, straight, and parallel conductors, separated by one meter in a vacuum, produces a force of exactly 2×10^{-7} newtons per meter of length between them.
- Kelvin (K):** The kelvin is the base unit for temperature. It is defined by utilizing the Boltzmann constant, which relates the average kinetic energy of particles in a system to its temperature.
- Mole (mol):** The mole is the base unit for amount of substance. It is defined as the amount of substance that contains as many entities (typically atoms or molecules) as there are in 0.012 kilograms of carbon-12.
- Candela (cd):** The candela is the base unit for luminous intensity. It measures luminous power per unit solid angle emitted by a light source in a particular direction.

SI BASE UNITS:

Physical Quantity	Basic Unit	Basic Unit Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous Intensity	candela	cd

SI Derived Units:

SI Derived Units are units of measurement formed by combining the seven base SI units in various mathematical ways. These units allow us to express a wide range of physical quantities in a standardized and consistent manner. Here are a few examples of SI derived units:

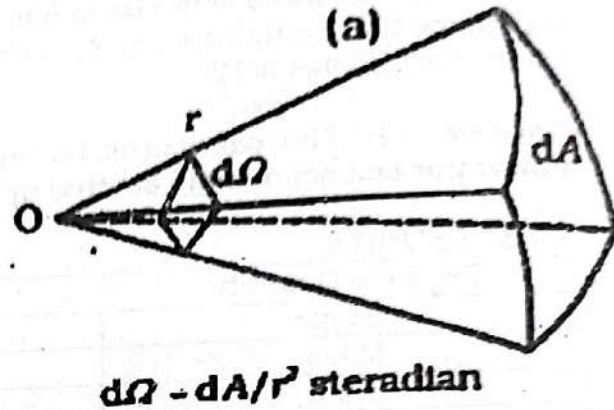
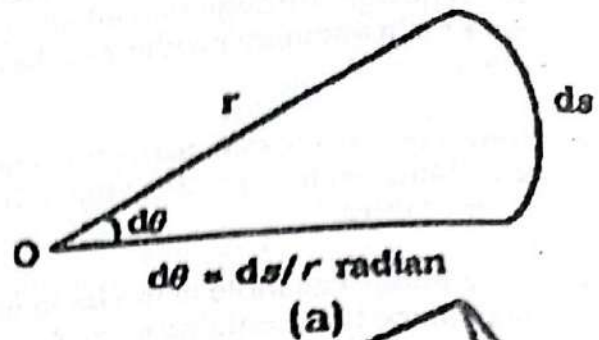
- Newton (N):** The newton is the unit of force. In the SI system and is defined as one kilogram-meter per second square ($1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$). It is derived from the base units of mass, length, and time.
- Joule (J):** The joule is the unit of energy (or work). It is defined as one newton-meter ($1 \text{ J} = 1 \text{ N} \cdot \text{m}$). It is derived from the units of force and length.
- Pascal (Pa):** The pascal is the unit of pressure and stress and is defined as one newton per square meter ($1 \text{ Pa} = 1 \text{ N/m}^2$). It is derived from the units of force and area.
- Watt (W):** The watt is the unit of power and is defined as one joule per second ($1 \text{ W} = 1 \text{ J/s}$). It is derived from the units of energy and time.
- Coulomb (C):** The coulomb is the unit of electric charge and is defined as the charge transported by one ampere in one second ($1 \text{ C} = 1 \text{ A} \cdot \text{s}$). It is derived from the units of electric current and time.
- Volt (V):** The volt is the unit of electric potential and electric potential difference and is defined as one joule per coulomb ($1 \text{ V} = 1 \text{ J/C}$). It is derived from the units of energy and electric charge.
- Hertz (Hz):** The hertz is the unit of frequency and is defined as one cycle per second ($1 \text{ Hz} = 1/\text{s}$). It is derived from the unit of time.

Supplementary Units:

Supplementary units also known as supplementary SI units or units of supplementary dimension, are a category of units used in the International System of Units (SI) to express quantities that are dimensionless. These units are not derived from the seven base SI units but are nevertheless crucial for describing certain physical properties and quantities.

Supplementary units include:

- **Radian (rad):** The radian is the unit used to measure angles in the SI system. It is defined as the angle subtended at the center of a circle by an arc whose length is equal to the radius of the circle. Since angles are dimensionless, the radian is a supplementary unit.
- **Steradian (sr):** The steradian is the unit used to measure solid angles in three-dimensional space. It is defined as the solid angle subtended at the center of a sphere by a surface area equal to the square of the sphere's radius. Like the radian, the steradian is dimensionless and is considered a supplementary unit.



CONVENTION AND STANDARDIZATION IN UNITS:

In physics, we follow set rules for units to keep our measurements clear and consistent. Here are some basic conventions for units in physics:

- **International System of Units (SI):** The SI system is the universal standard for measuring units in physics. It provides a set of basic and derived units for various physical quantities, ensuring clear and precise communication in scientific measurements.
- **Base Units:** The SI system has seven base units that form the basis for measuring fundamental physical quantities.
- **Utilization of Prefixes:** The SI system uses prefixes like kilo, mega, milli, micro, and nano to show multiples or fractions of base units. These prefixes make it easier to express measurements across a wide range of magnitudes. Common SI prefixes include kilo (k), mega (M), milli (m), micro (μ), and nano (n).
- **Maintaining Consistent Formatting:** Consistent formatting of units is crucial in academic writing. Units are generally written in lowercase, with exceptions for symbols derived from scientists' names, such as the Kelvin unit (K). Additionally, the singular form is customary for units, except when quantities exceed one, as in '10 meters'.
- **Appropriate Unit Selection:** Choosing the right units is crucial in academic work. It prevents confusion and ensures accurate representation of physical quantities. For example, using meters per second (m/s) for velocity and meters per second squared (m/s^2) for acceleration is a good practice.

Dimensions

DIMENSIONS:

A dimension is a fundamental property used to describe and categorize physical quantities. It is typically associated with a specific unit of measurement and represents a distinct aspect of a quantity. The dimensions of a physical quantity are the power to which the units of the base quantity are raised to represent a derived unit of that quantity. OR
Dimensions are representation of a physical quantity in terms of Mass[M], Length[L] and Time[T].

There are seven fundamental dimensions in terms of which the dimensions of all other physical quantities may be expressed.

- **Length [L]:** Length is a dimension associated with measurements of size and distance. Its SI unit is the meter (m). For example, when measuring the length of a bookshelf, you are using the dimension of length.
- **Mass [M]:** Mass is a dimension that characterizes the amount of matter in an object. Its SI unit is the kilogram (kg). When measuring the mass of a car, you are working with the dimension of mass.
- **Time [T]:** Time is a dimension related to the measurement of events' durations or intervals. Its SI unit is the second (s). Timing a race or calculating the duration of a movie involves the dimension of time.
- **Electric Current [I]:** Electric current is a dimension that quantifies the flow of electric charge through a conductor. Its SI unit is the ampere (A). When measuring the current in an electrical circuit, you are working with the dimension of electric current.
- **Temperature [K]:** Temperature is a dimension representing the hotness or coldness of an object. Its SI unit is the kelvin (K). Checking the temperature with a thermometer involves the dimension of temperature.
- **Amount of Substance [N]:** Amount of substance is a dimension that measures the quantity of entities (atoms, molecules, ions, etc.) in a sample. Its SI unit is the mole (mol). When determining the number of molecules in a chemical reaction, you are dealing with the dimension of amount of substance.
- **Luminous Intensity (Iv):** Luminous Intensity is a dimension characterizing the brightness of light. Its SI unit is the candela (cd). When assessing the brightness of a light source, you are considering the dimension of luminous intensity.

Application of Dimensionality:

- To validate the consistency of dimensional equation
- To derive relationship between different physical quantities
- To convert unit of a physical quantity
- To prove the homogeneity of Equation

(For Examples See Short Question Answer(Q3,Q4,Q6) and Extra Numericals)

Limitation of Dimensionality:

- Doesn't provide information about constant
- Formula containing trigonometric functions, exponential and log functions cannot be derived alone.
- Dimension does not indicate whether the quantity is scalar or vector.

Chapter 1

Errors and Uncertainty

ERRORS:

Error in the context of measurement and scientific analysis refers to the discrepancy between a measured or observed value and the true or expected value of a quantity

Errors can be divide into two types:

1. Systematic Errors:

Systematic errors are a type of error in measurements that consistently shift the results in one direction, either higher or lower, from the true value or target. These errors are often the result of flaws or biases in the measurement system or procedure and can lead to inaccuracies that persist across multiple measurements. Systematic errors are also known as determinate errors because they have a consistent and determinable effect on the measurements.

Various type of systematic errors are

Instrumental Error:	Instrumental errors occur when the equipment or measuring instrument used in an experiment malfunctions or operates inaccurately, leading to measurement errors.
Environmental Error:	Environmental errors arise when external factors, such as the conditions in the laboratory or surroundings, introduce inaccuracies into the experiment
Observational Error:	Observational errors occur when a scientist makes inaccuracies while reading or recording measurements.
Theoretical Error:	Theoretical errors manifest when the theoretical model or assumptions used in an experiment do not align with reality, leading to inaccurate results

2. Random Errors:

Random errors, also known as indeterminate errors, are fluctuations or variations in measured values that occur due to unpredictable and uncontrollable factors. These errors are inherent to the measurement process and arise from a combination of numerous factors, including inherent instrument limitations, environmental conditions, and human limitations in making precise measurements.

Various type of random errors are

Environmental Error:	Environmental errors occur due to variations in the external conditions surrounding the measurement process. These variations can affect the measurement instrument or the conditions under which the measurements are taken. Examples of environmental errors include: • Temperature Fluctuations • Humidity Variations • Vibration effects • EM Interference
Observational Error:	Observational errors stem from mistakes or limitations in the human process of making measurements or recording data. These errors can result from factors related to perception, judgment, or technique. Examples of observational errors include: • Parallax Errors • Misalignment • Judgmental Error • Data entry Error

UNCERTAINTY:

In any experiment, we make several measurements with a certain level of precision. These measurements are never completely precise, and there's always a bit of uncertainty involved. You can think of uncertainty as the gap between the measurement we actually take (which can be affected by the equipment or methods we use) and the standard or expected value. It's important to note that uncertainties should not be confused with errors.

Calculation of Uncertainty:

Example: Fractional Uncertainty:

Suppose you are measuring the time it takes for a pendulum to complete one full swing, and you obtain a value of 2.45 seconds. However, you know that your stopwatch has an uncertainty of ± 0.05 seconds due to its precision.

To calculate the fractional uncertainty, you divide the uncertainty by the measured value:

$$\text{Fractional Uncertainty} = \frac{(\text{Uncertainty})}{(\text{Measured Value})}$$

$$\text{Fractional Uncertainty} = \frac{(0.05 \text{ seconds})}{(2.45 \text{ seconds})} = 0.0204$$

So, the fractional uncertainty in this measurement is approximately 0.0204

Example: Percentage Uncertainty

Using the same pendulum experiment, you can express the uncertainty as a percentage of the measured value:

$$\begin{aligned} \text{Percentage Uncertainty} &= (\text{Fractional Uncertainty}) \times 100\% \\ \text{Percentage Uncertainty} &= (0.0204) \times 100\% = 2.04\% \end{aligned}$$

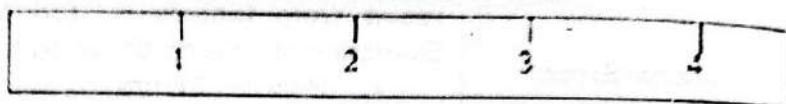
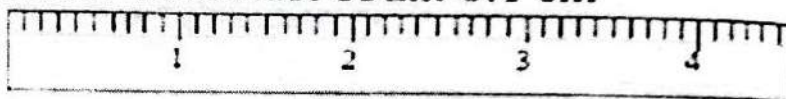
The percentage uncertainty in this measurement is 2.04%.

LEAST COUNT OR RESOLUTION:

The least count of a measuring instrument is the smallest change that can be measured with the instrument. It is the smallest division on the scale of the instrument. For example, if the least count of a ruler is 0.1 cm, then the smallest change that can be measured with the ruler is 0.1 cm.

The least count of a measuring instrument determines the precision of the instrument. A high precision instrument has a smaller least count and can therefore measure smaller changes.

For example, if you use a ruler with a least count of 0.1 cm to measure the length of a pencil, and you get a reading of 10.2 cm, then you can be sure that the pencil is at least 10.2 cm long. However, you cannot be sure that the pencil is exactly 10.2 cm long, because the least count of the ruler is 0.1 cm. The pencil could be anywhere between 10.2 cm and 10.3 cm long.

Ruler with least count 1 cm**Ruler with least count 0.1 cm****ACCURACY VS PRECISION:**

Accuracy is how close a measurement is to the true value of the quantity being measured.

Precision is how close repeated measurements are to each other.

Another way to think about it is that accuracy is the "closeness to the target" and precision is the "repeatability" of the measurement.

It is possible to be accurate but not precise, and vice versa.

For example, if you throw a dart at a target and it hits close to the edge, it is accurate but not precise. If you throw multiple darts at the target and they are all scattered around, the darts are precise but not accurate.

In general, it is better to have both accuracy and precision. However, in some cases, it may be more important to have one than the other. For example, if you are measuring the distance to a planet, it is more important to have accurate measurements than precise measurements.

Significance of Resolution:

Using an instrument of smallest resolution is important for several reasons:

Reasons	Explanation
Precision	Precision is the reproducibility of a measurement, and it is important to use an instrument with the smallest resolution possible to ensure maximum precision and consistency.
Accuracy	Precision is important for accuracy because it ensures that measurements are as close as possible to their true values.
Detail	Precision allows for the measurement of finer details and features, which is important in various applications.
Reduced uncertainty:	Precision reduces uncertainty, which makes measurements more accurate.

GRAPHS:

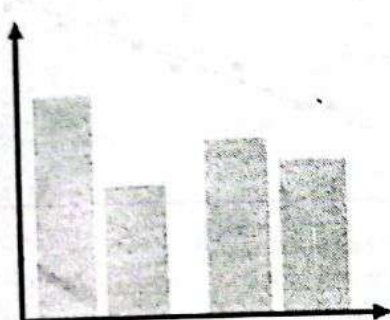
Graphs are visual representations that illustrate relationships between data sets. They are used to display data in an easily understandable and memorable way, highlighting trends, patterns, and relationships.

Dependent and Independent Variables:

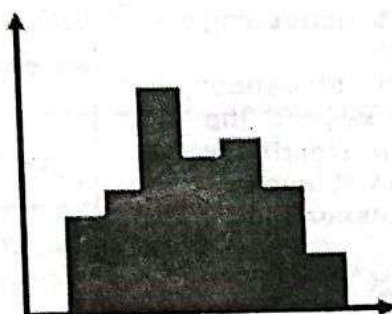
In statistics and mathematical modelling, two terms are used to describe the relationship between variables. The independent variable is manipulated to observe its effect on the dependent variable, which is measured or observed and expected to change due to the manipulation. Typically, the independent variable is plotted on the x-axis, while the dependent variable is plotted on the y-axis in graphical representations.

Types of Graphs: Various types of graphs, such as scatter plots, line graphs, and bar graphs, can be used to represent the relationship between dependent and independent variables. The choice of graph depends on the data type and the nature of the relationship.

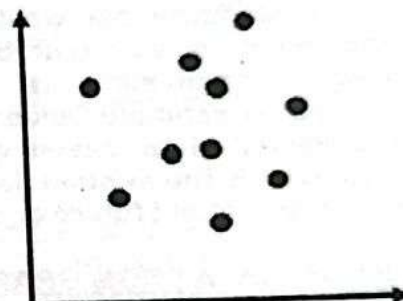
Bar chart



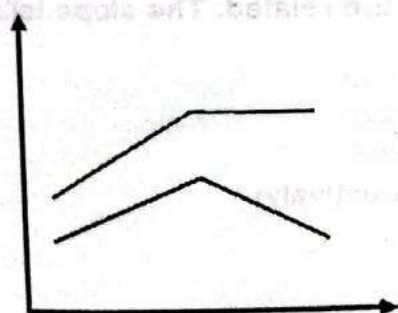
Histogram



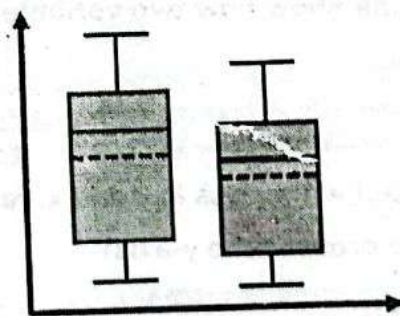
Scatter plot



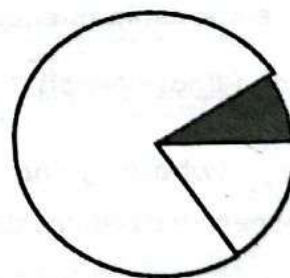
Line chart



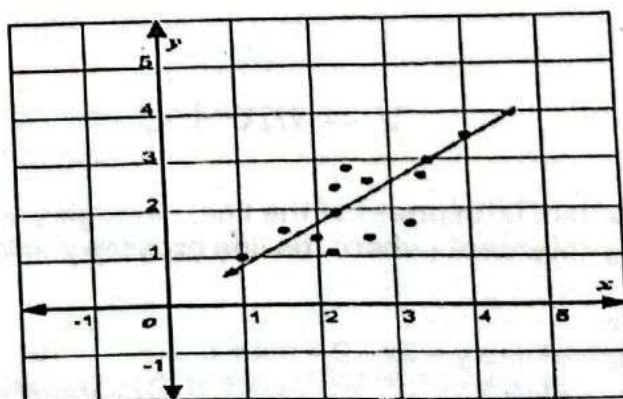
Boxplot



Pie chart



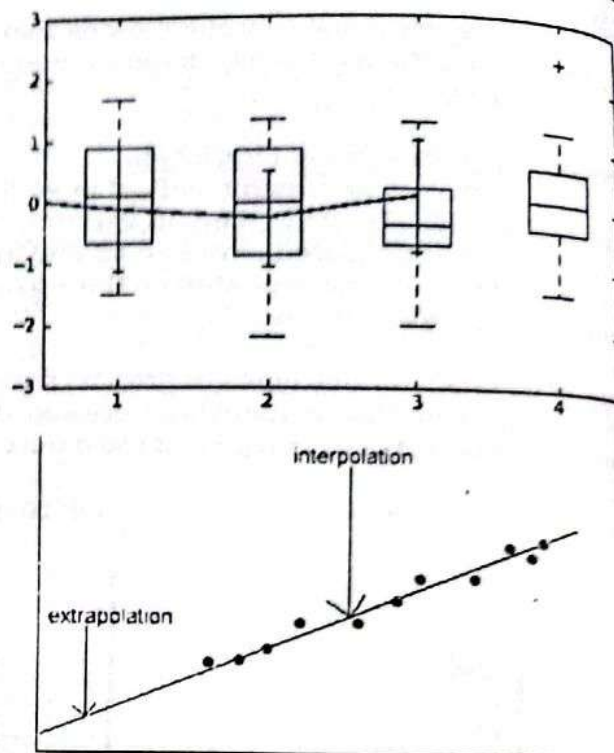
Best Fit Line Graph: A best fit line graph visually displays the relationship between two variables, showing the general trend in the data. The line of best fit is drawn to represent the underlying pattern in the data.



Error Bars: Error bars are used in graphs to represent uncertainty or variability in data. Shorter error bars indicate greater precision, while longer error bars indicate less precision. They help assess the reliability and accuracy of data and determine the statistical significance of differences between groups or experiments.

Interpolation: Interpolation is like filling in the gaps between known points. It's a way to estimate values within the range of data we already have by looking at the patterns in that data. It helps us figure out what's happening between the points we've measured.

Extrapolation: Extrapolation is a statistical technique used to estimate values beyond the range of collected data based on trends or patterns observed in the existing data. It involves making predictions about future or unseen data.



INTERCEPTING DATA FROM GRAPH

Linear and nonlinear graphs are graphical representations of mathematical relationships between variables. The **slope** of a line or curve represents the **rate of change** of one variable with respect to another. The **Intercept** is the point where the line or curve crosses the **x-axis** or **y-axis**. In short, linear and nonlinear graphs show how two variables are related. The **slope** tells you how quickly the relationship changes.

Linear and non-linear graph

In linear graph

Slope = $m = \frac{\Delta y}{\Delta x}$ (where Δy and Δx are the changes in y and x , respectively)

- Intercept = c (where the line crosses the y -axis)

The general equation of straight line is:

$$y = mx + c$$

Where,

m is gradient (steepness of the line)

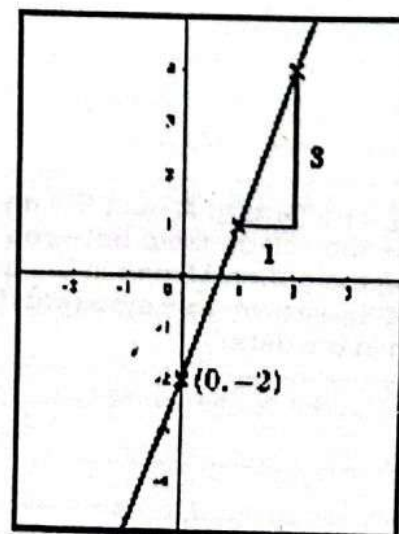
c is the y -intercept (where the line crosses y -axis)

Example:

The graph of line $y = 3x - 2 = mx + c$

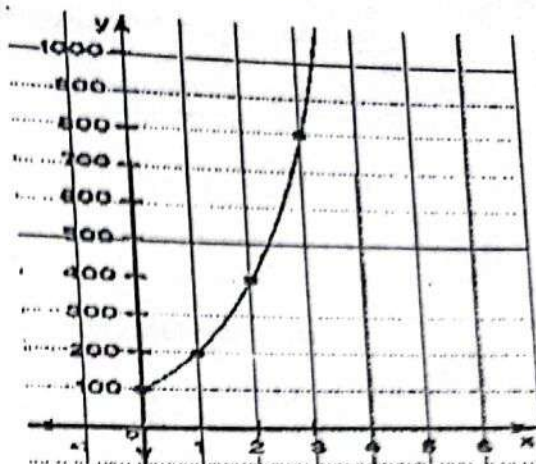
The gradient is 3

The y -intercept is -2, the coordinates (0, -2)



In non-linear graph

- Slope and intercept cannot be calculated using the same formulas as linear graphs
- The relationship between the variables is more complex
- The graph may have multiple intercepts, or no intercept at all



Chapter 1

Significant Figures, Rounding off Data & Exponential Notations

DEFINITIONS:

Significant Figures

Significant figures are the number of digits in a value, often a measurement, that contribute to the degree of accuracy of the value.
or
The digits in a number show reliability in measurement

Rounding off Data

The process to reduce a number up to desired significant figures & adjust the last reported digit

Exponential Notations

Exponential notation is a way of writing numbers that are too large or too small to be conveniently written in decimal form. It is a shorthand way of writing repeated multiplication. It consists of following parts.

$$a \times 10^n$$

- a = coefficient = can be any number, except for 0.
- n = exponent = is any integer, including negative integers.

SIGNIFICANT FIGURES:

Rules:

- Non-zero digits are significant figures

Example:

253 has 3 significant figures

3.29 has 3 significant figures

- Zero lying between non-zero numbers are significant figures

Example:

5005 has 4 significant figures

50103 has 5 significant figures

- Zero just after a decimal point in a number less than 1 is not significant

Example:

0.00536 has 3 significant figures

0.00008 has 1 significant figure

- Zero after a decimal point in a number greater than 1 is significant

Example

4.0000 has 5 significant figures

2.000 has 4 significant figures

- The final zero in large numbers does not have a decimal is not significant

Example

1000 has 1 significant figure

123500 has 4 significant figure

ROUNDING OFF DATA:

Rules:

- (a) If the final digit is greater than 5 then drop it by adding 1 to the second last digit.

Example

2.256 round off to 2.26 (3 significant digits)

6.28 round off to 6.3 (2 significant digits)

- (b) If the final digit is less than 5 then simply drop it without adding 1 in the second last digit

Example

3.952 round off to 3.95 (3 significant digits)

6.23 round off to 6.2 (2 significant digits)

- (c) If the final digit is 5 then:

- If the second last digit is even (i.e. 2, 4, 6, 8) then simply drop 5

Example

4.865 are rounded off to 4.86 (3 significant digits)

6.825 are rounded off to 6.82 (3 significant digits)

- if the second last digit is odd (i.e. 1, 3, 5, 7, 9) then add 1 to the second last digit for making it odd and drop 5

Example

4.835 are rounded off to 4.84 (3 significant digits)

6.875 are rounded off to 6.88 (3 significant digits)

EXPONENTIAL NOTATION:

Representation:

Exponential Notation can be generally represented as,

$$a \times 10^n$$

where a is co-efficient & it is a number between 1 to 9.99
 n is the exponent & it is an integer raised to base 10

if the number is greater than one then the exponent is positive
 if the number is lesser than one then the exponent is negative

INTER CONVERSION OF STANDARD & EXPONENTIAL NOTATION:

Standard Notation Into Exponential Notation

Rules:

- The decimal point moves towards the left if the value is greater than 10
- The decimal point moves towards the left if the value is greater than 10

Example:

Standard Notation		Exponential Notation
400000000	=	4.0×10^8
0.000045	=	4.5×10^{-5}

Exponential Notation Into Standard Notation

Rules

- The decimal point moves towards the right for large numbers with a positive exponent
- The decimal point moves towards the left for small numbers with a negative exponent

Exponential Notation		Standard Notation
4.0×10^8	=	400000000
4.5×10^{-5}	=	0.000045

APPLICATIONS OF EXPONENTIAL NOTATION:

Addition & Subtraction

Rule:

- All numbers should be converted to the same exponent

Example	Solution
Addition of 2.03×10^3 4.05×10^2	2.030×10^3 0.405×10^3 2.435×10^3

Multiplication

- Multiply all digits (coefficient)
- Add all Exponents
- Adjust the decimal just after 1st non zero digits if not present

Example	Solution
Multiplication of 1.02×10^3 2.03×10^2	1.02×10^3 2.03×10^2 $2.071 \times 10^{3+2}$ 2.071×10^5

Division

- Divide all digits (coefficient)
- Subtract the Exponents of the denominator from the numerator
- Adjust the decimal just after 1st non zero digits if not present

Example	Solution
Division of 2.03×10^3 with 4.05×10^2	2.03×10^3 4.05×10^2 $0.50 \times 10^{3-2}$ 0.50×10^1 5.0×10^0 or 5.0

Physics XI

Powers

- Apply power to digits (coefficient)
- Multiply Power & Exponents
- Adjust the decimal just after 1st non zero digits if not present

Example
Solve
 $(2.03 \times 10^3)^2$

Solution
 $(2.03 \times 10^3)^2$
 $(2.03)^2 \times 10^{3 \times 2}$
 4.12×10^6

Roots

- adjust the decimal so that exponents are divisible by root to get the whole number
- Apply root to digits (coefficient)
- Divide Exponents by Root
- Adjust the decimal just after 1st non zero digits if not present

Example
Solve
 $\sqrt{2.03 \times 10^3}$

Solution
 $(0.203 \times 10^4)^{\frac{1}{2}}$
 $(0.203)^{\frac{1}{2}} \times 10^{\frac{4}{2}}$
 0.45×10^2
 4.5×10^1
or 45

PRACTICE QUESTIONS:

S#	Problem
1	Write the significant figures of following (a) 300 (b) 13.06 (c) 0.2653 (d) 562000 (e) 032.2560 (f) 208 (g) 0.0045 (h) 10.08
2	Reduce 5.36251958 to 1 significant figure & 4 significant figures
3	How many significant figures are there in 936.120 convert into 2 & 3 significant figure
4	Simplify the following by using exponential notation $5320 + 1923 + 1205$
5	Express the following as a power of 10 (a) 5678 (b) 432.1 (c) 2000 (d) 54321 (e) 500000 (f) 53.24 (g) 0.000005

Chapter 1

Multiple Choice Questions

MCQ'S	ANS.
1. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^3 are: a) 5, 1, 2 b) 5, 1, 5 c) 5, 5, 2 d) 4, 4, 2	a
2. Which among the following is the supplementary unit: a) Mass b) Time c) Solid angle d) Luminosity	c

3. The unit of solid angle is a) second c) kilogram	b) Steradian d) Candela	b
4. The quantity having the same unit in all system of unit is: a) Mass c) Length	b) Time d) Temperature	b
5. Random errors can be eliminated by: a) taking number of observations and their mean b) measuring the quantity with more than one instrument c) eliminate the cause d) careful observations		a
6. Systemic error can be: a) either positive or negative c) negative only	b) positive only d) zero error	a
7. [MLT ⁻²] is dimensional formula of: a) strain c) displacement	b) force d) pressure	b
8. Which of the following pair has the same dimension? a) moment of inertia and torque c) surface tension and force	b) impulse and momentum d) specific heat and latent heat	b
9. Dependent Variable is a) cause c) effect	b) cause and effect d) reason	c
10. Dimension of Kinetic Energy is same as that of: a) Acceleration c) velocity	b) work d) force	b

***** Exams Practice MCQs *****

1. What is the slope of the line represented by the equation $y = 2x + 1$? (a) 2 (b) 1 (c) 0 (d) Infinity	a
2. Which of the following is the definition of accuracy? (a) The closeness of a measurement to the true value. (b) The closeness of repeated measurements to each other. (c) Both (a) and (b). (d) None of the above.	a
3. Which of the following is the definition of precision? (a) The closeness of a measurement to the true value. (b) The closeness of repeated measurements to each other. (c) Both (a) and (b). (d) None of the above.	b
4. Which of the following is an example of a high-precision instrument? (a) A ruler (b) A stopwatch (c) A balance (d) All of the above.	c
5. Which of the following is an example of a high-accuracy instrument? (a) A ruler (b) A stopwatch (c) A balance (d) All of the above.	c

Physics XI

6. What is resolution? (a) The smallest change in a quantity that can be measured or detected. (b) The ability of an instrument to distinguish between two closely spaced objects. (c) Both (a) and (b). (d) None of the above.	c
7. Why is resolution important? (a) It allows us to make more accurate measurements. (b) It allows us to see smaller details. (c) Both (a) and (b). (d) None of the above.	c
8. Which of the following has a higher resolution? (a) A digital camera with 10 megapixels or a digital camera with 20 megapixels? (b) A microscope with a magnification of 100x or a microscope with a magnification of 1000x? (c) A ruler with a scale of 1 cm or a ruler with a scale of 1 mm? (d) All of the above.	d
9. Which of the following is not a benefit of high resolution? (a) Improved accuracy of measurements. (b) Ability to see smaller details. (c) Reduced noise in images and videos. (d) Increased file size.	d
10. What is the significance of resolution in the field of medicine? (a) High-resolution medical imaging devices allow doctors to diagnose diseases more accurately. (b) High-resolution medical imaging devices allow doctors to perform surgery more precisely. (c) Both (a) and (b). (d) None of the above.	c
11. How many significant figures does the number 12.345 have? (a) 1 (b) 2 (c) 3 (d) 4	d
12. How many significant figures does the number 0.00345 have? (a) 1 (b) 2 (c) 3 (d) 4	b
13. What is the correct answer to the following calculation, rounded to the correct number of significant figures? $123.4 + 56.78 = 180.18$ (a) 180 (b) 180.1 (c) 180.18 (d) 180.2	d
14. Which of the following numbers has the same number of significant figures as the number 1.2345? (a) 123.45 (b) 0.12345 (c) 12.3450 (d) 1.234500	c
15. Which of the following is true about significant figures? (a) Significant figures are the digits in a number that are reliable and meaningful. (b) Significant figures include all non-zero digits and any zeros between them. (c) Significant figures include all non-zero digits and any trailing zeros. (d) All of the above.	d
16. What is extrapolation? (a) The process of estimating a value outside the range of the known data. (b) The process of estimating a value inside the range of the known data.	a

(c) The process of finding the best fit line through a set of data points. (d) None of the above	
17. Which of the following is a derived quantity? (a) Length (b) Mass (c) Time (d) Velocity	d
18. What is systematic error? (a) An error that is consistent in one direction, either positive or negative. (b) An error that is unpredictable and varies randomly. (c) An error that is caused by human error. (d) An error that is caused by a faulty instrument.	a
19. What is random error? (a) An error that is consistent in one direction, either positive or negative. (b) An error that is unpredictable and varies randomly. (c) An error that is caused by human error. (d) An error that is caused by a faulty instrument.	b
20. Which of the following is an example of systematic error? (a) A ruler that is too short. (b) A balance that is not level. (c) A stopwatch that is inaccurate. (d) All of the above.	d
21. Which of the following is an example of random error? (a) Reading the temperature from a thermometer at a different angle each time. (b) Measuring the mass of an object on a balance that is not level. (c) Using a stopwatch that is inaccurate. (d) All of the above.	a
22. What is the difference between error and uncertainty? (a) Error is the difference between the measured value and the true value, while uncertainty is the range of possible values that the true value could be within. (b) Error is the uncertainty in a measurement, while uncertainty is the error in a measurement. (c) Error and uncertainty are the same thing. (d) None of the above.	a
23. What is the best way to estimate uncertainty? (a) By taking multiple measurements and calculating the standard deviation. (b) By using a variety of instruments. (c) By calibrating instruments regularly. (d) None of the above	a
24. What is the purpose of dimensional analysis? (a) To check the consistency of equations. (b) To derive new equations. (c) To estimate the values of physical quantities. (d) All of the above	d
25. Which of the following measuring instruments is used to measure time (a) Stopwatch (b) Clock (c) Pendulum (d) All of the above	d

26. Which of the following measuring instruments is used to measure the length of a straight line? (a) Ruler (b) Vernier calipers (c) Micrometer screw gauge (d) All of the above	a
27. Which of the following is a type of measuring instrument used to measure electrical quantities? (a) Ammeter (b) Voltmeter (c) Ohmmeter (d) All of the above	d
28. Which of the following measuring instruments is used to measure current? (a) Ammeter (b) Voltmeter (c) Ohmmeter (d) All of the above	a
29. Which of the following measuring instruments is used to measure voltage? (a) Ammeter (b) Voltmeter (c) Ohmmeter (d) All of the above	b
30. Which of the following measuring instruments is used to measure resistance? (a) Ammeter (b) Voltmeter (c) Ohmmeter (d) All of the above	c
31. Which of the following is a base unit in the SI system? (a) Meter (b) Kilogram (c) Second (d) All of the above	d
32. Which of the following is a derived unit? (a) Velocity (b) Force (c) Energy (d) All of the above	d
33. Which of the following is a supplementary unit? (a) Radian (b) Steradian (c) Both (a) and (b) (d) None of the above	c
34. Which branch of physics studies the motion of objects and the forces that act on them? (a) Mechanics (b) Thermodynamics (c) Electromagnetism (d) Quantum mechanics	a
35. Which branch of physics studies the relationship between heat and other forms of energy? (a) Mechanics (b) Thermodynamics (c) Electromagnetism (d) Quantum mechanics	b
36. Which branch of physics studies the interaction of electric and magnetic fields? (a) Mechanics (b) Thermodynamics (c) Electromagnetism (d) Quantum mechanics	c
37. Which branch of physics studies the behavior of matter at the atomic and subatomic level? (a) Mechanics (b) Thermodynamics (c) Electromagnetism (d) Quantum mechanics	d
38. Which branch of physics studies the behavior of light and other forms of electromagnetic radiation? (a) Optics (b) Acoustics (c) Nuclear physics (d) Particle physics	a
39. Which branch of physics studies the behavior of sound waves? (a) Optics (b) Acoustics (c) Nuclear physics (d) Particle physics	b
40. Which branch of physics studies the structure and behavior of nuclei? (a) Optics (b) Acoustics (c) Nuclear physics (d) Particle physics	c

Percent Uncertainty	$\frac{\text{Absolute uncertainty}}{\text{measured value}} \times 100$
Uncertainty in Addition	$\text{Let } x_1 = A \pm \Delta A$ $x_2 = B \pm \Delta B$ $x = x_1 + x_2$ $x = (A \pm \Delta A) + (B \pm \Delta B)$ $x = (A + B) \pm (\Delta A + \Delta B)$
Uncertainty in Difference	$x = x_1 - x_2$ $x = (A - B) \pm (\Delta A + \Delta B)$
Uncertainty in Multiplication	$x = x_1 \times x_2$ $x = (A \times B) \pm (\Delta A\% + \Delta B\%)$
Uncertainty in Division	$x = x_1 \div x_2$ $x = (A \div B) \pm (\Delta A\% + \Delta B\%)$
Uncertainty in Power factor	$x = (A \pm \Delta A)^n$ $x = A^n \pm n\Delta A\%$
Error	Measured Value - True Value
Uncertainty	$\frac{1}{2} \times \text{least count}$
measurement	best Estimate $\pm$ uncertainty